Customer Analytics Dashboard Documentation

Project Overview

The Customer Analytics Dashboard is an end-to-end analytics project designed to analyze customer behavior, revenue trends, purchasing activity, and customer segmentation using PostgreSQL and Tableau.

This project simulates a real-world Business Intelligence workflow where raw data is transformed into meaningful business insights through SQL analytics and interactive dashboards.

Project Objectives

The primary goals of this project were:

- Analyze customer purchasing behavior
- Measure business revenue performance
- Identify top-performing customer segments
- Track sales growth trends over time
- Understand customer engagement using recency analysis
- Build an interactive executive-level analytics dashboard

Technology Stack

Component	Technology
Database	PostgreSQL
Query Language	SQL
Visualization	Tableau
Data Modeling	Star Schema Concepts
Analytics	KPI & Customer Analytics
Data Warehousing Concepts	Bronze, Silver, Gold Layers

Data Warehousing Concepts Applied

This project follows layered data warehousing concepts:

Bronze Layer

Raw source data ingestion.

Purpose:

- Store raw CSV/source data
- Maintain traceability
- Preserve source records

Implemented Concepts:

- Full data loads
- Raw transactional ingestion
- No transformations

Silver Layer

Data cleaning and transformation layer.

Purpose:

- Standardize data
- Handle missing values
- Create derived metrics
- Prepare analytical datasets

Implemented Concepts:

- Data cleansing
- Data standardization
- Derived columns
- Data enrichment

Gold Layer

Business-ready analytical layer.

Purpose:

- Build reporting-ready datasets
- Aggregate business metrics
- Support dashboards and analytics

Implemented Concepts:

- Aggregated measures
- KPI-ready tables

- Customer analytics views

ETL & Data Loading Workflow

The project simulates ETL workflows using SQL procedures.

Implemented operations:

- Table truncation
- Bulk CSV loading
- Data ingestion logging
- Batch execution handling
- Error handling

Example:

- Bronze layer loading using bulk insert procedures
- Source CRM and ERP data integration

Dashboard Overview

The dashboard contains:

1. KPI Cards
2. Sales Trend Analysis
3. Revenue by Customer Segment
4. Revenue by Age Segment
5. Customer Recency Analysis
6. Top Customers Analysis

KPI Section

KPI 1 — Total Sales

What It Measures

Total revenue generated from all customer purchases.

Business Question Answered

“How much total revenue did the business generate?”

Value Observed

$29.36M

Business Insight

The company generated approximately $29 million in total sales, indicating strong business activity and customer purchasing power.

KPI 2 — Average Order Value

What It Measures

Average revenue generated per order.

Business Question Answered

“How much does a customer spend on average per transaction?”

Value Observed

~911.7

Business Insight

Customers place relatively high-value orders, indicating healthy transaction size and purchasing behavior.

KPI 3 — Total Customers

What It Measures

Total unique customers in the dataset.

Business Question Answered

“How many customers does the business serve?”

Value Observed

18,484

Business Insight

The business serves a large customer base, allowing segmentation and behavioral analysis.

KPI 4 — Total Orders

What It Measures

Total number of orders placed.

Business Question Answered

“How many total transactions occurred?”

Value Observed

27,659

Business Insight

The order volume indicates active customer engagement and repeat purchasing activity.

Visualization Analysis

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1. Sales Trend Over Time

Chart Type

Line Chart

Purpose

Visualize revenue growth across months and years.

Business Questions Answered

- Is revenue increasing over time?
- Which periods show rapid growth?
- Are there seasonal trends?

Key Insights

- Revenue remained relatively stable initially
- Significant growth started during later periods
- Strong upward trend observed in 2013
- Indicates business expansion and improved customer acquisition

Business Value

Helps executives:

- Forecast revenue
- Monitor growth trends
- Identify peak business periods

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2. Revenue by Customer Segment

Chart Type

Horizontal Bar Chart

Customer Segments

- NEW
- REGULAR
- VIP

Purpose

Compare revenue contribution by customer groups.

Business Questions Answered

- Which customer segment generates the highest revenue?
- Which segment should receive business focus?

Key Insights

- NEW customers contributed the highest revenue
- REGULAR customers contributed moderate revenue
- VIP contribution was comparatively smaller

Business Interpretation

The company is successfully acquiring new customers.

Possible interpretation:

- Strong acquisition campaigns
- High first-time purchase behavior
- Opportunity to improve customer retention strategies

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3. Revenue by Age Segment

Chart Type

Horizontal Bar Chart

Purpose

Analyze revenue contribution across age groups.

Business Questions Answered

- Which age group generates the most revenue?
- Which demographic is most valuable?

Key Insights

- Customers aged 50+ generated the highest revenue
- Age group 40–49 also contributed strongly
- Younger age groups contributed significantly less

Business Interpretation

Older customer demographics possess higher purchasing power and stronger revenue contribution.

Possible business action:

- Target premium marketing campaigns toward older customers

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4. Customer Recency Analysis

Chart Type

Histogram

Purpose

Analyze customer engagement based on recency.

What Recency Means

Recency measures how recently customers made purchases.

Lower recency:

- More recently active customers

Higher recency:

- Less recently active customers

Business Questions Answered

- Are customers actively purchasing?
- How engaged is the customer base?
- Are customers becoming inactive?

Key Insights

- Majority of customers fall within lower recency ranges
- Indicates strong recent engagement
- Very few customers show high recency values

Business Interpretation

Most customers are actively interacting with the business, indicating healthy customer retention and engagement.

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5. Top 10 Customers by Revenue

Chart Type

Horizontal Bar Chart

Purpose

Identify highest-value customers.

Business Questions Answered

- Who are the top revenue-generating customers?
- Which customers contribute most to sales?

Key Insights

- A small group of customers contributes substantial revenue
- Revenue concentration exists among top buyers

Business Value

Helps business teams:

- Create loyalty programs
- Build VIP engagement strategies
- Improve customer retention for high-value customers

Dashboard Design Features

Implemented dashboard enhancements:

- KPI cards
- Consistent color palette
- Responsive layout
- Professional spacing
- Business-friendly formatting
- Interactive dashboard structure
- Executive-level presentation

Real-World Skills Demonstrated

This project demonstrates practical skills in:

SQL Skills

- Aggregations
- Grouping
- Filtering
- Analytical calculations
- KPI engineering

Tableau Skills

- Dashboard design
- KPI cards
- Histograms
- Bar charts
- Line charts
- Formatting & layout optimization

Data Engineering Concepts

- ETL workflow understanding
- Data warehouse layering
- Business-ready data modeling

• Data integration concepts

Business Intelligence Skills

• Storytelling with data
• Business interpretation
• Executive reporting
• Customer analytics

Business Impact of the Dashboard

This dashboard enables businesses to:

• Monitor overall business health
• Track revenue performance
• Understand customer behavior
• Identify profitable customer groups
• Improve retention strategies
• Support executive decision-making

Future Enhancements

Planned future improvements:

• Product Analytics Dashboard
• Executive Dashboard
• Interactive Filters
• Drill-down analysis
• Predictive analytics
• Customer Lifetime Value (CLV)
• Cohort analysis

Conclusion

The Customer Analytics Dashboard successfully transforms raw customer transaction data into actionable business insights.

The project demonstrates:

• End-to-end analytics workflow
• Data warehouse understanding
• SQL analytics
• Tableau dashboarding
• Business intelligence storytelling

This project closely simulates real-world analytics solutions used in modern organizations for customer and revenue analysis.